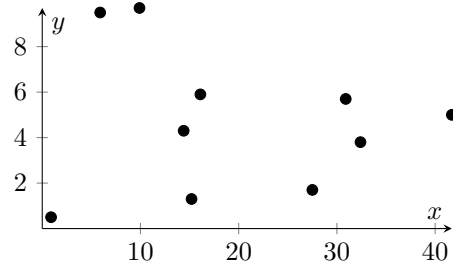
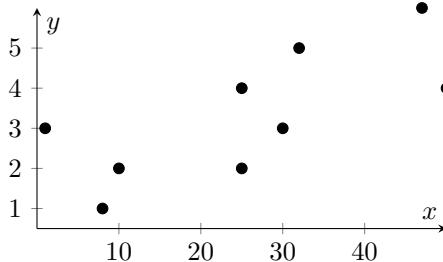


Correlation and Regression - Solutions



Exercise 5.1. The data for the first two plots in this booklet is given in the tables below. Calculate the covariance for each, and comment on what it means.

X_1	Y_1
1	3
10	2
25	4
32	5
47	6
8	1
30	3
25	2
50	4

First, calculate the means:

$$\bar{x} = \frac{1 + 10 + 25 + \dots + 50}{9} = \frac{228}{9} = 25.33$$

$$\bar{y} = \frac{3 + 2 + 4 + \dots + 4}{9} = \frac{30}{9} = 3.33$$

$$\begin{aligned} \text{cov}(X, Y) &= \frac{(1 - 25.33)(3 - 3.33) + (10 - 25.33)(2 - 3.33) + \dots + (50 - 25.33)(4 - 3.33)}{9} \\ &= \frac{(-24.33 \times -0.33) + (-15.33 \times -1.33) + \dots + (24.67 \times 0.67)}{8} = \frac{153}{8} \\ &= 19.125 \end{aligned}$$

Since this is positive, it indicates a positive correlation between the two variables.

X_2	Y_2
32.4	3.8
0.9	0.5
30.9	5.7
14.4	4.3
5.9	9.5
41.7	5
15.2	1.3
27.5	1.7
9.9	9.7
16.1	5.9

Calculate the means:

$$\bar{x} = \frac{32.4 + 0.9 + 30.9 + \dots + 16.1}{10} = \frac{194.9}{10} = 19.49$$

$$\bar{y} = \frac{3.8 + 0.5 + 5.7 + \dots + 5.9}{9} = \frac{47.4}{10} = 4.74$$

$$\begin{aligned} \text{cov}(X, Y) &= \frac{(32.4 - 19.49)(3.8 - 4.74) + (0.9 - 19.49)(0.5 - 4.74) + \dots + (16.1 - 19.49)(5.9 - 4.74)}{9} \\ &= \frac{(12.91 \times -0.94) + (-18.59 \times -4.24) + \dots + (-3.39 \times 1.16)}{9} = \frac{-40.126}{9} \\ &= -4.45 \end{aligned}$$

Since this is negative, it indicates a negative correlation between the two variables.

Exercise 5.2. For the two sets of points in Exercise 5.1, calculate the PMCC.

Calculate the standard deviations:

$$\sigma_{X_1} = \sqrt{\frac{1}{n-1} \left(\sum x^2 - \frac{(\sum x)^2}{n} \right)} = \sqrt{\frac{1}{8} \left(8048 - \frac{(228)^2}{9} \right)} = \sqrt{284} = 16.85$$

$$\sigma_{X_2} = \sqrt{\frac{1}{8} \left(120 - \frac{(30)^2}{9} \right)} = \sqrt{2.5} = 1.58$$

$$\sigma_{Y_1} = \sqrt{\frac{1}{9} \left(5330.95 - \frac{(194.9)^2}{10} \right)} = \sqrt{170.261} = 13.05$$

$$\sigma_{Y_2} = \sqrt{\frac{1}{9} \left(314.4 - \frac{(47.4)^2}{10} \right)} = \sqrt{9.97} = 3.16$$

Then we can calculate the PMCC:

$$\rho_1 = \frac{\text{cov}(X_1, Y_1)}{\sigma_{X_1} \sigma_{Y_1}} = \frac{19.125}{16.85 \times 1.58} = 0.718$$

This shows a very strong positive correlation.

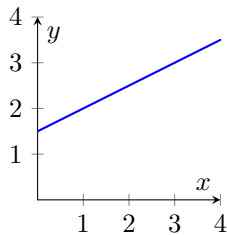
$$\rho_2 = \frac{\text{cov}(X_2, Y_2)}{\sigma_{X_2} \sigma_{Y_2}} = \frac{19.125}{13.05 \times 3.16} = -0.108$$

This shows a weak negative correlation.

Exercise 5.3. Write down the equation for a line with each of the following properties, and then sketch the line.

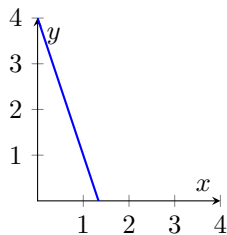
- (a) The line has a slope of 0.5 and a y-intercept of 1.5

$$y = 1.5 + 0.5x$$



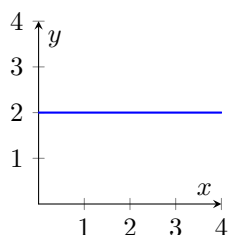
- (b) The line has a slope of -3 and a y-intercept of 4

$$y = 4 - 3x$$



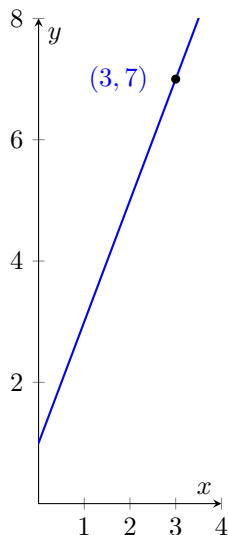
- (c) The line is horizontal and crosses the y-axis at 2

$$y = 2 + 0x, \text{ or } y = 2$$



(d) The line starts at $(0,1)$ and goes to the point $(3,7)$

Since $y = 1$ when $x = 0$, the y -intercept is 1. The line goes across 3 and up 6, so its slope is $\frac{6}{3} = 2$. Then $y = 1 + 2x$



Exercise 5.4. For the regression lines and points given below, calculate the residual.

(a) $y = 5 + 2.8x$; $(-1.5, 2)$

The point on the line for which $x = -1.5$ has y -coordinate

$$y = 5 + (2.8 \times -1.5) = 0.8$$

The difference between this and the point given is

$$2 - 0.8 = 1.2$$

(b) $y = 3.1 + 1.5x$; $(4, 1)$

The point on the line for which $x = 4$ has y -coordinate

$$y = 3.1 + (1.5 \times 4) = 9.1$$

The difference between this and the point given is

$$1 - 9.1 = -8.1$$

(c) $y = 1 + 0.5x$; $(1.5, 3)$

The point on the line for which $x = 1.5$ has y -coordinate

$$y = 1 + (0.5 \times 1.5) = 1.75$$

The difference between this and the point given is

$$3 - 1.75 = 1.25$$

Exercise 5.5. For the data sets given below, find the equation of the least squares regression line.

(a)

X	1	2	3	4
Y	4	1	9	6

$$\bar{x} = 2.5, \bar{y} = 5$$

(b)

X	9	9	4	4
Y	9	10	7	4

$$\bar{x} = 6.5, \bar{y} = 7.5$$

(c)

X	8	6	6	8
Y	6	4	9	4

$$\bar{x} = 7, \bar{y} = 5.75$$

(a)

$$\begin{aligned} b &= \frac{\sum(x - \bar{x})(y - \bar{y})}{\sum(x - \bar{x})^2} \\ &= \frac{(1 - 2.5)(4 - 5) + (2 - 2.5)(1 - 5) + (3 - 2.5)(9 - 5) + (4 - 2.5)(6 - 5)}{(1 - 2.5)^2 + (2 - 2.5)^2 + (3 - 2.5)^2 + (4 - 2.5)^2} \\ &= \frac{7}{5} = 1.4 \end{aligned}$$

$$a = \bar{y} - b\bar{x} = 5 - (1.4 \times 2.5) = 4.375$$

So the line is $y = 4.375 + 1.4x$

(b)

$$\begin{aligned} b &= \frac{\sum(x - \bar{x})(y - \bar{y})}{\sum(x - \bar{x})^2} \\ &= \frac{(9 - 6.5)(9 - 7.5) + (9 - 6.5)(10 - 7.5) + (4 - 6.5)(7 - 7.5) + (4 - 6.5)(4 - 7.5)}{(9 - 6.5)^2 + (9 - 6.5)^2 + (4 - 6.5)^2 + (4 - 6.5)^2} \\ &= \frac{20}{25} = 0.8 \end{aligned}$$

$$a = \bar{y} - b\bar{x} = 7.5 - (0.8 \times 6.5) = 2.3$$

So the line is $y = 2.3 + 0.8x$

(c)

$$\begin{aligned} b &= \frac{\sum(x - \bar{x})(y - \bar{y})}{\sum(x - \bar{x})^2} \\ &= \frac{(8 - 7)(6 - 5.75) + (6 - 7)(4 - 5.75) + (6 - 7)(9 - 5.75) + (8 - 7)(4 - 5.75)}{(8 - 7)^2 + (6 - 7)^2 + (6 - 7)^2 + (8 - 7)^2} \\ &= \frac{-3}{4} = -0.75 \end{aligned}$$

$$a = \bar{y} - b\bar{x} = 5.75 - (-0.75 \times 7) = 11$$

So the line is $y = 11 - 0.75x$

Exercise 5.6. For the datasets given in Exercise 5.5, predict what the output would be for the input value 5.

(a) *If $x = 5$ then y can be found as:*

$$y = 4.375 + 1.4x = 4.375 + (1.4 \times 5) = 11.375$$

(b) *If $x = 5$ then y can be found as:*

$$y = 2.3 + 0.8x = 2.3 + (0.8 \times 5) = 6.3$$

(c) *If $x = 5$ then y can be found as:*

$$y = 11 - 0.75x = 11 + (-0.75 \times 5) = 7.25$$